

CAMBRIDGE BAY, NU, Canada

WMO: 712880

Lat: 69.107N

Lon: 105.116W

Elev: 19

StdP: 101.10

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-41.8	-40.4	-45.3	0.0	-41.4	-44.4	0.0	-40.3	15.9	-26.4	14.1	-27.7	3.6	40	1.106

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	17.6	12.9	15.5	11.6	13.5	10.5	13.2	17.0	11.9	15.0	10.7	13.3	5.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.1	8.2	14.6	9.9	7.6	13.5	8.7	7.0	12.3	37.3	17.1	33.9	15.1	30.9	13.5	18.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.3	10.7	DB	-43.2	21.2	2.0	2.2	-44.7	22.8	-45.9	24.1	-47.0	25.3	-48.4	27.0	(4)
(5)			WB	-43.1	15.8	1.8	1.6	-44.4	16.9	-45.4	17.9	-46.4	18.8	-47.7	19.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.0	-30.9	-31.9	-28.6	-20.3	-8.8	3.0	9.3	7.4	0.8	-9.2	-20.6	-27.6	(6)	
	DBStd	15.68	5.62	5.44	6.19	5.87	5.56	3.91	3.13	3.19	3.70	5.70	6.10	5.99	(7)	
	HDD10.0	8446	1268	1172	1197	909	583	214	53	92	276	597	917	1167	(8)	
	HDD18.3	11442	1527	1406	1455	1159	842	461	281	338	526	855	1167	1426	(9)	
	CDD10.0	46	0	0	0	0	0	3	31	13	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	Wind	WSAvg	5.4	5.3	5.2	5.1	5.3	5.2	5.2	5.1	5.3	5.8	6.0	5.5	5.3	(14)
	Precipitation	PrecAvg	150	5	5	6	8	11	14	27	28	21	14	7	5	(15)
(16) Precipitation	PrecMax	232	12	16	25	25	30	44	84	78	47	42	17	16	(16)	
	PrecMin	92	0	0	0	0	0	1	3	8	5	2	1	0	(17)	
	PrecStd	36	3	4	6	7	7	11	20	15	9	8	4	4	(18)	
	PrecStd	36	3	4	6	7	7	11	20	15	9	8	4	4	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-12.3	-14.5	-9.7	-3.8	3.0	17.6	21.6	19.6	10.9	2.1	-3.2	-8.6	(19)	
		MCWB	-12.6	-15.0	-10.3	-4.3	1.8	12.6	15.3	14.1	8.6	1.6	-3.6	-9.1	(20)	
	2%	DB	-17.2	-17.5	-12.8	-6.2	1.6	13.6	18.5	16.6	8.6	0.8	-7.0	-12.7	(21)	
		MCWB	-17.6	-17.9	-13.3	-6.8	0.8	10.0	13.3	12.7	7.2	0.4	-7.3	-13.1	(22)	
	5%	DB	-20.0	-20.2	-15.8	-8.4	0.5	10.8	16.6	14.1	7.1	-0.3	-9.0	-15.3	(23)	
		MCWB	-20.4	-20.5	-16.3	-8.9	0.0	8.0	12.1	11.0	5.9	-0.7	-9.3	-15.6	(24)	
	10%	DB	-22.3	-23.4	-18.9	-11.3	-0.8	8.5	14.9	12.2	5.8	-1.5	-11.4	-18.6	(25)	
		MCWB	-22.6	-23.7	-19.4	-11.8	-1.4	6.3	11.1	9.8	4.9	-1.9	-11.8	-19.0	(26)	
	0.4%	WB	-12.5	-14.9	-10.2	-4.3	2.2	12.9	15.7	14.8	8.9	1.8	-3.5	-9.0	(27)	
		MCDB	-12.2	-14.5	-9.6	-3.9	2.7	17.4	20.4	18.4	10.4	2.0	-3.2	-8.7	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-17.5	-17.9	-13.2	-6.8	1.0	10.2	13.7	12.9	7.4	0.5	-7.3	-13.1	(29)	
		MCDB	-17.2	-17.5	-12.8	-6.3	1.4	13.3	17.7	16.1	8.4	0.7	-7.0	-12.8	(30)	
	5%	WB	-20.3	-20.5	-16.3	-8.9	0.0	8.1	12.5	11.4	6.1	-0.6	-9.3	-15.7	(31)	
		MCDB	-20.0	-20.2	-15.8	-8.5	0.5	10.6	16.0	13.7	7.0	-0.3	-9.0	-15.3	(32)	
	10%	WB	-22.6	-23.7	-19.3	-11.8	-1.5	6.4	11.3	10.0	5.0	-1.9	-11.7	-19.0	(33)	
		MCDB	-22.3	-23.4	-18.9	-11.4	-0.8	8.4	14.5	11.8	5.7	-1.5	-11.4	-18.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	6.4	7.1	8.1	6.7	5.4	7.1	5.6	3.9	5.3	6.6	6.3	(35)	
		MCDBR	9.2	9.4	8.7	8.1	5.5	9.7	10.3	9.1	5.8	4.4	7.4	9.0	(36)	
		MCWBR	9.1	9.3	8.5	8.1	5.2	6.7	6.4	5.7	4.5	4.6	7.4	9.0	(37)	
	5% WB	MCDBR	9.2	9.5	8.6	8.1	5.4	9.6	9.9	8.5	5.5	4.5	7.4	8.9	(38)	
		MCWBR	9.1	9.4	8.5	8.1	5.2	6.8	6.7	5.8	4.5	4.6	7.4	8.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.146	0.210	0.243	0.292	0.310	0.353	0.358	0.337	0.300	0.233	0.161	N/A	(40)	
	taud		1.898	2.043	2.221	2.130	2.172	2.284	2.374	2.461	2.471	2.140	1.888	N/A	(41)	
	Ebn at Noon		77	569	800	844	871	839	822	802	738	519	42	0	(42)	
	Edh at Noon		4	43	71	108	117	110	97	80	59	38	4	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.02	0.42	1.97	4.37	6.26	5.92	5.17	3.35	1.61	0.49	0.06	0.00	(44)	
	RadStd		0.00	0.03	0.10	0.19	0.26	0.39	0.51	0.27	0.17	0.09	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page